

An Analysis on the Significance of References and Varied Representation in the Engineering Design Process

Introduction:

This analysis discusses the development of my understanding of the engineering design process by reflecting on how two key observations of teamwork changed my interpretation of Praxis ideas. I first considered the impact that a lack of early reference designs had on framing our design brief, then explored how multiple perspectives can create a variety of representations. I also discuss how I hope to apply this enriched understanding to future experiences.

Argument 1: The Importance of Reference Designs in Framing

Reflecting on the process of developing the design brief with my team, revealed the importance of reference designs as a tool for guiding thinking in the design processes.

I observed that my group was creative, but would try to develop concepts before ensuring they were viable first. While planning our brief, we accidentally came up with solutions rather than illustrating the design possibility. We delayed the process of researching and analysing reference designs until we had committed to our ideas (figure 1).

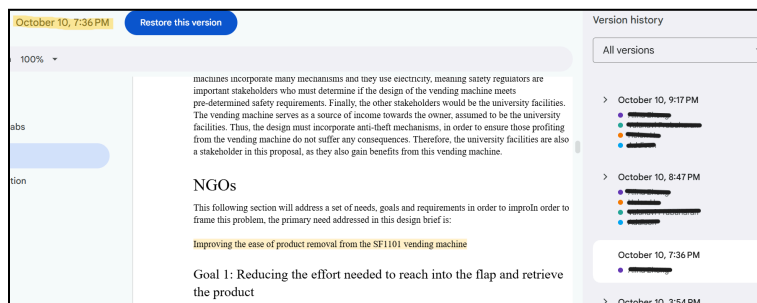


Figure 1: Our design brief the day before it was due, missing a section for reference designs. [1]

Because of this, we did not have a unanimous understanding of our references (figure 2), which led the communication of our framing to be unclear. We did not adequately demonstrate how current devices could be applied to our opportunity and why they fail to address our issue. Our feedback (figure 3 & 4) highlighted how the references fell short. Having references guide our thinking earlier may have helped us all be on the same page about our design space and better frame and justify the opportunity throughout.

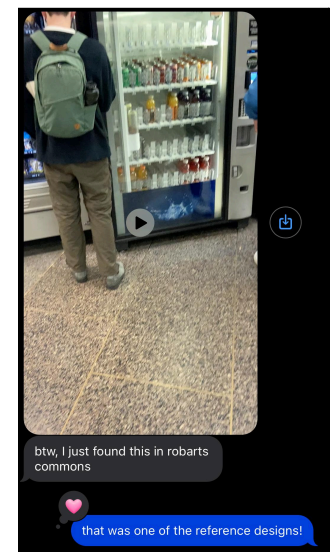


Figure 2: Messages from October 22, depicting members with a lack of knowledge about reference designs. [2][3]

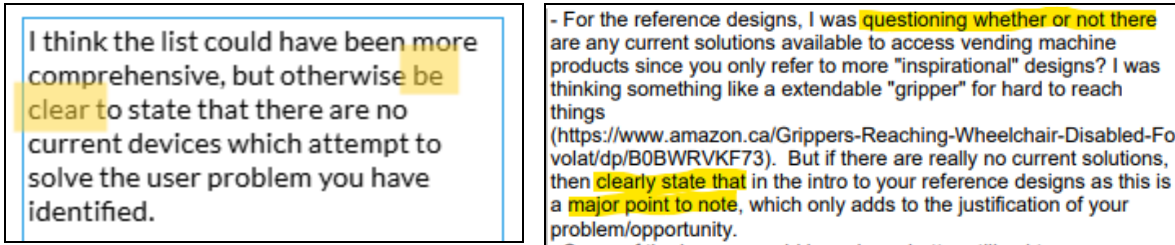


Figure 3, 4: Samples of feedback received on reference designs. [4]

We struggled to understand the value of references for effective framing and justification within our design brief. Primary and secondary research developed our NGOs but we researched to find metrics for ideas we already had, instead of using it to develop ideas. We were unaware of how impactful reference designs could be when defining and justifying our opportunity. We failed to fully embrace a growth mindset and became fixated on thinking our initial ideas were best, rather than using other's ideas to help us iterate and refine. This enhanced my interpretation of the concept of a 'growth mindset'. Reflecting on this helped me realize one cannot expect new information to come to them. I assumed the growth comes from being told new information and embracing it by critical thinking and developing ideas. Through observing the impacts of our oversight, I have deepened my understanding to include the necessity of seeking information that challenges preconceived notions. Now I understand the engineering design process as something where you must continuously seek out what others have done before you. I also think it is imperative to the process to continuously challenge your ideas.

I intend on embracing this enriched growth mindset in future engineering processes and daily life by taking initiative and looking for transformative information. I hope to use it to gain new perspectives that impact how I frame and justify future opportunities. I plan on starting design processes with secondary research and seek references early to help form ideas.

Argument 2: The Impact of Representing Ideas in a Variety of Ways

Reflecting on conversations during the diverging process for Alpha, I now believe it is valuable to have multiple representations of the same idea.

I observed my teammates prefer to demonstrate their ideas in a different way. While discussing a confusing n+1 solution when working with a morph chart, we represented a single idea through multiple mediums: verbal descriptions, physical demonstration with a low fidelity prototype and gestures (figure 5), sketches (figure 6), and analogies and references (figure 7). I preferred a combination of methods and therefore "translated" between methods to help other teammates interpret.

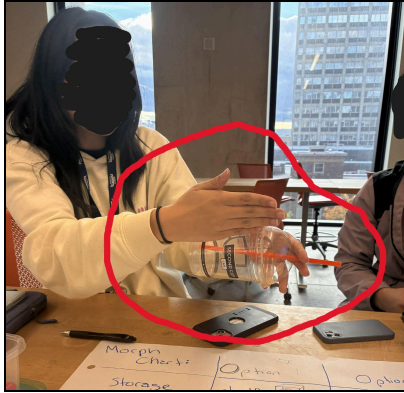


Figure 5: Demonstration with coffee cup and hands gestures.

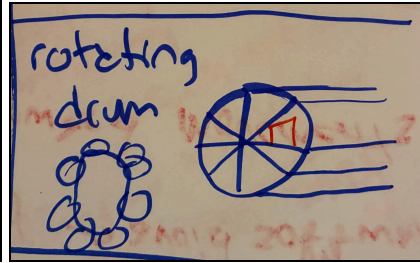


Figure 6: Drawings of what solution could look like. [5]

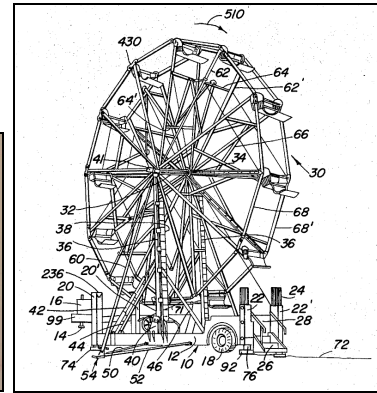


Figure 7: Ferris wheel used as an analogy. [6]

Having various depictions of this idea helped our group gain shared understanding. We were effective in communicating with each other, and easily able to explain it to people less familiar with our project. A strong team dynamic developed and we could build off of each other's ideas from the common foundation.

Representing one idea in numerous ways allowed us to consider aspects of the design initially overlooked. We rapidly iterated and adjusted our claim using Toulmin's model effectively, leading us to quickly recognize this idea did not fit in our scope or meet our requirements. Therefore, it did not make it through the next round of converging. However, the conversation stuck with me for its demonstration of how worldview and identity can affect perception, interpretation, and representation. This deepened my understanding of the Praxis 'reality and theory' model [7]. This conversation exemplifies how everyone perceived the solution differently and therefore modelled their understanding differently. Although we used different vessels, we created a shared understanding of the world as we dream it - our potential solution. Previously, I only appreciated the connection between the world we dream and create, but analyzing this process of representation highlighted another important part of the model. I value how we perceive and model the world more. I understand the engineering design process does not just come from following procedures and creating ideas for specific circumstances, but is built on each engineer's interpretation of the opportunity, solution, and best representation. It is an important balance between having a shared understanding of the task, and diversifying interpretations to be more holistic.

I will be more intentional in interpreting and representing phases of the design process. I want to ensure each team member can develop their own ideas before we discuss so individual perspectives have the opportunity to form and continue to develop a safe space for diverse perspectives in all future groups I work with.

Conclusion

This analysis transformed my perspective of the early engineering design process. I will carry more intention, utilizing a growth mindset and embracing multiple perspectives.

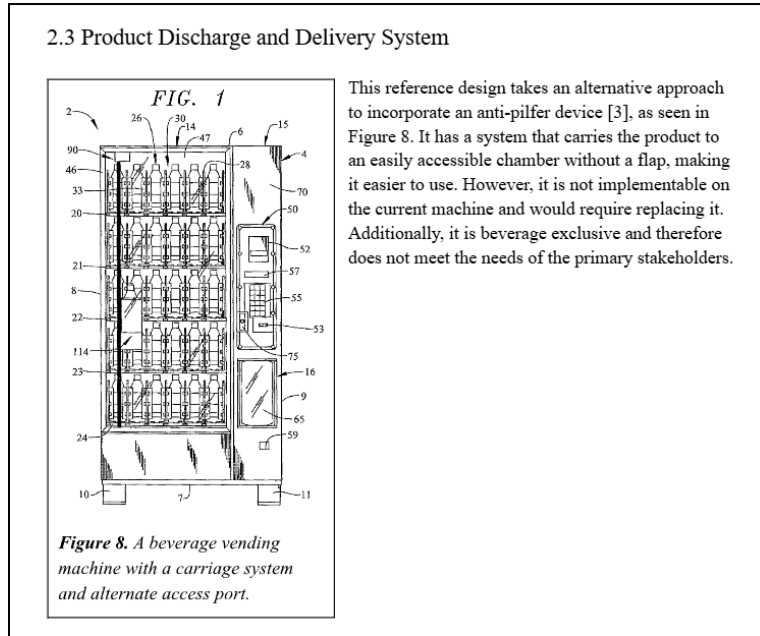
Appendix:

[1] The full design brief imaged. The only work done for references is highlighted yellow.

☰ Design Brief - October 10

[2] The full video that was sent in the group chat [here](#).

[3] The section depicting that reference design in the design brief:



And its citation: J. P. Ardern, D. Bowen, P. Miller, C. W. Percy, K. A. Poyck, and D. Schmidt, "Product discharge and delivery system for a vending machine," Jan. 17, 2012 Accessed: Oct. 10, 2025. [Online]. Available:

[https://patents.google.com/patent/US8096444B2/en?q=\(Vending+Machine+retrieval\)&oq=Vending+Machine+retrieval](https://patents.google.com/patent/US8096444B2/en?q=(Vending+Machine+retrieval)&oq=Vending+Machine+retrieval)

[4] Full screenshots to demonstrate how reference designs were the majority of our constructive feedback

	Key things that detracted from the quality of your brief
I think the list could have been more comprehensive, but otherwise be clear to state that there are no current devices which attempt to solve the user problem you have identified. I was thinking something like a "gripper" for hard to reach things, but specific for vending machines: https://www.amazon.ca/Grippers-Reaching-Wheelchair-Disabled-Fovolat/dp/B0BWRVKF73 I agree with [redacted] comment here. I think a greater variety of existing vending machines and some type of retrieval mechanism would be helpful as reference designs.	- The stakeholder section could have been better formatted in a list or table to enhance readability. Moreover, I think the list of primary users could have been more comprehensive to take into account not only faculty, staff, and visitors of SF, but also people with mobility issues, such as wheelchair users. This may have been outside your design scope, so fine, but the Stakeholder section is used to assess particular individuals/groups who may be impacted by your design. - For the reference designs, I was questioning whether or not there are any current solutions available to access vending machine products since you only refer to more "inspirational" designs? I was thinking something like a extendable "gripper" for hard to reach things (https://www.amazon.ca/Grippers-Reaching-Wheelchair-Disabled-Fovolat/dp/B0BWRVKF73). But if there are really no current solutions, then clearly state that in the intro to your reference designs as this is a major point to note, which only adds to the justification of your problem/opportunity. - Some of the images could have been better utilized to communicate the intended idea, particularly the reference design figures.

[5] The full morph chart, drawing from figure in the top right.

Morph Chart:

	Option 1	Option 2	Option 3	Option 4
Storage	shelf 	bin / box 	hanging 	rotating drum 
dispensing	dropper 	conveyor device (elevator) 	claw 	slide 
retrieving	robot automated sensor  spring magnet 	flip 	drawer 	grate / open 
payment	Cash 	tap 	facial recognition (Amazon, wing) 	trade 
Selection	button 	audible 	eye tracking 	
packaging	boxes 	bags 	bottles 	batch (cereal dispenser) 

[6] Citation of patent image came from: W. C Deem, "Portable folding ferris wheel," Jan. 05, 1971

Available: <https://patents.google.com/patent/US3552747A/en>

Please note the patent is expired but this is not the particular one she researched, instead she references a ferris wheel in general and this serves as a representation of that for this analysis.

[7] Image of said model:

